

Self-Introduction & Research Plan at HCMUT

Keisuke Asami

GD1, Science of Technology Innovation

Theory of Machine Learning Laboratory (TML Lab)

Nagaoka University of Technology

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About Me

Keisuke Asami (浅見 圭祐)

Program:

- 5-year Integrated Doctoral Program, Science of Technology Innovation (GD1, equivalent to M1)

University:

- Nagaoka University of Technology

Lab:

- Theory of Machine Learning Laboratory

Supervisor:

- Assoc. Prof. Gendo Kumoi

Research Interest

- Automatic extraction of personal decision-making styles from behavioral histories using LLMs

Hobbies

- Visiting cafés
- Traveling

I'm really excited to explore Vietnam!

Our Laboratory

Theory of Machine Learning Laboratory (TML Lab)

University: Nagaoka University of Technology

Supervisor: Assoc. Prof. Gendo Kumoi

Our lab conducts research across a wide range of fields.

Among these, one line of research — which I'm working on:

Automatically extracting individuals' ****decision-making styles**** from behavioral histories as natural language documents, and integrating them with recommender systems and generative AI.

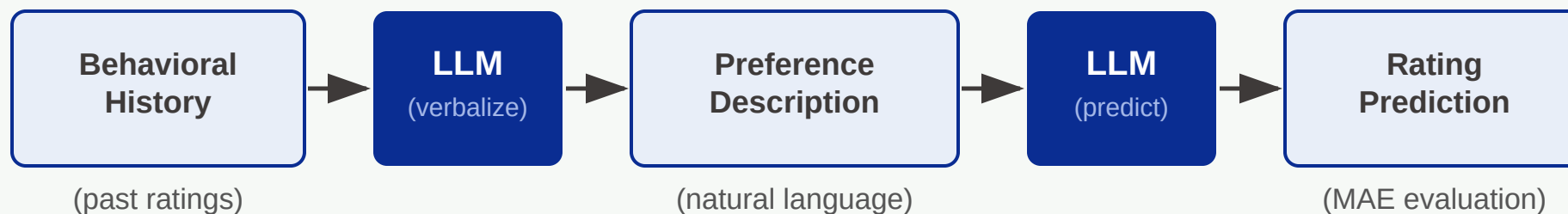
Existing collaboration with HCMUT:

- Joint papers with Châu Nguyễn Hồng, Thảo Nguyễn Thanh, Tiên Quách Mỹ

Background: Existing Research

Framework: Preference Verbalization via LLM

Core idea: Use LLM-generated preference descriptions to predict future ratings. Prediction accuracy (MAE) serves as an objective measure of verbalization quality.



Results (SITA2025 / IEICE2026):

- Preference descriptions improve LLM rating prediction accuracy
- Effectiveness varies by user characteristics

Open question: Does this framework generalize across different languages?

My Research at HCMUT

Research Topic

Automatic Extraction of Decision-Making Styles from Behavioral Histories under Japanese-Vietnamese × L1/L2 Cross-Lingual Conditions

Core Question: When the same behavioral history is interpreted in different languages, do the extracted decision-making styles remain consistent?

In other words: Does a **language-invariant preference representation** exist? This is a fundamental design question for multilingual recommender systems.

Methodology

2 × 2 Cross-Lingual Consistency Matrix

	Style extracted in L1	Style extracted in L2
AI Interview in L1	① Same-language (baseline)	③ Cross-lingual
AI Interview in L2	④ Cross-lingual	② Same-language (baseline)

- **Cells ① & ②:**
 - Same-language consistency (baseline)
- **Cells ③ & ④:**
 - Cross-lingual consistency → reveals **language-invariant** preference

Similarity measure: Multilingual Sentence Embeddings

Participants: N=10 Japanese (Japan) + N=10 Vietnamese (HCMUT)

Research Plan

3-Phase Research Plan

Phase	Period	Location	Key Activities
Phase 1	– Aug 2026	Japan (NUT)	Japanese data (N=10); model implementation; baseline
Phase 2	Sep – Nov 2026	Vietnam (HCMUT)	Vietnamese data (N=10); 2×2 matrix experiments
Phase 3	Dec 2026 –	Japan (NUT)	Integrated analysis (N=20); paper writing; OSS release

Expected outputs:

- Japanese-Vietnamese $\times$ L1/L2 corpus (open on GitHub)
- Cross-lingual consistency evaluation framework (OSS)
- International co-authored paper (RecSys / EMNLP Workshop, etc.)

Collaboration Request

What We Need from HCMUT

1. **Participant Recruitment** — ~N=10 Vietnamese participants per study
 - Behavioral history data (video viewing / product purchases)
 - AI interviews in Vietnamese (L1) + English (L2)
2. **Your Support** — We warmly welcome any collaboration and support from Prof. Tho's laboratory in making this research a success.

We are truly looking forward to working with Prof. Tho's laboratory.
Thank you very much for your time!

Thank You for Listening!

Keisuke Asami

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